

Scalar Field

```
if (pba->has_fld == _TRUE_) {

    /* get rho_fld from vector of integrated variables */
    pvecback[pba->index_bg_rho_fld] = pvecback_B[pba->index_bi_rho_fld];

    /* get w_fld from dedicated function */
    class_call(background_w_fld(pba,a,&w_fld,&dw_over_da,&integral_fld), pba->error_message, pba->error_message);
    pvecback[pba->index_bg_w_fld] = w_fld;

    // Obsolete: at the beginning, we had here the analytic integral solution corresponding to the case  $w=w_0+w_1(1-a/a_0)$ :
    // pvecback[pba->index_bg_rho_fld] = pba->Omega0_fld * pow(pba->H0,2) / pow(a,3.*(1.+pba->w0_fld+pba->wa_fld)) * exp(3.*pba->wa_fld*(a-1.));
    // But now everthing is integrated numerically for a given w_fld(a) defined in the function background_w_fld.

    rho_tot += pvecback[pba->index_bg_rho_fld];
    p_tot += w_fld * pvecback[pba->index_bg_rho_fld];
    dp_dloga += (a*dw_over_da-3*(1+w_fld)*w_fld)*pvecback[pba->index_bg_rho_fld];
}
```

```
/* Derivative of total pressure w.r.t. conformal time */
pvecback[pba->index_bg_p_tot_prime] = a*pvecback[pba->index_bg_H]*dp_dloga;
if (pba->has_scf == _TRUE_) {
    /** The contribution of scf was not added to dp_dloga, add p_scf_prime here: */
    pvecback[pba->index_bg_p_prime_scf] = pvecback[pba->index_bg_phi_prime_scf]*|
        (-pvecback[pba->index_bg_phi_prime_scf]*pvecback[pba->index_bg_H]/a-2./3.*pvecback[pba->index_bg_dV_scf]);
    pvecback[pba->index_bg_p_tot_prime] += pvecback[pba->index_bg_p_prime_scf];
}
```